Cessna 172S Skyhawk (C172S)

Lycoming IO-360-L2A, 180 hp · 4 seats · Typical 172S (1998+) POH procedures; fuel-injected engine. G1000 (2005+) and analogue-panel wording given where they differ.

V-speeds & limits

Vr	**55 KIAS**
Vx	**62 KIAS**
Vy	**74 KIAS**
Best glide	**68 KIAS (2,550 lb)**
Vs / Vso	**48 / 40 KIAS**
Vfe	**110 KIAS (10°) / 85 KIAS (10°–30°)**
Va	**105 KIAS (2,550 lb) / 98 KIAS (2,200 lb) / 90 KIAS (1,900 lb)**
Vno	**129 KIAS**
Vne	**163 KIAS**
Approach	**65–75 KIAS flaps up / 60–70 KIAS flaps 30° / 61 KIAS short field**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Fuel quantity	**CHECK visually (53 gal usable std)**
Oil quantity	**CHECK (8 qt max, 5 qt min)**
Passenger briefing	**COMPLETE (belts, doors, exits, sterile cockpit)**
Seats, belts, harnesses	**ADJUST & LOCK**
Brakes	**TEST & SET**
Control lock	**REMOVE**
Circuit breakers	**CHECK IN**
Electrical equipment / autopilot	**OFF**
Avionics switch (BUS 1 & BUS 2 on G1000)	**OFF**
Fuel selector	**BOTH**
Fuel shutoff valve	**ON (push full in)**
Standby battery	**TEST (hold 20 s, green light steady) then ARM (G1000)**
Master switch (ALT & BAT)	**ON**
Annunciators / PFD	**CHECK (LOW VOLTS on is normal before start)**

Engine Start

Throttle	**OPEN 1/4 inch**
Mixture	**IDLE CUTOFF**
Beacon	**ON**
Propeller area	**CLEAR**
Fuel pump	**ON**
Mixture	**RICH until fuel flow stable (3–5 s), then IDLE CUTOFF**
Fuel pump	**OFF**
Ignition switch	**START (release when engine fires)**
Mixture	**ADVANCE smoothly to RICH when engine fires**
Throttle	**ADJUST 1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60 s cold)**
Ammeter / LOW VOLTS annunciator	**CHECK charging / light out**
Avionics switch (BUS 1 & BUS 2 on G1000)	**ON**
Nav / strobe lights	**AS REQUIRED**

Before Taxi

Altimeter / barometric setting	**SET**
Heading indicator (analogue)	**SET to compass**
Radios / transponder / GPS	**SET**
Flaps	**CHECK operation, then UP**
Taxi light	**ON (as required)**
Parking brake	**RELEASE**
Brakes	**CHECK during taxi**
Flight instruments	**CHECK during turns (HSI / DG, turn coordinator, compass)**

Before Takeoff

Parking brake	**SET**
Seats, belts, harnesses	**CHECK SECURE**
Cabin doors	**CLOSED & LATCHED**
Flight controls	**FREE & CORRECT**
Flight instruments / PFD	**CHECK & SET (no red X)**
Fuel quantity	**CHECK**
Fuel selector	**BOTH**
Mixture	**RICH (below 3,000 ft)**
Elevator trim	**TAKEOFF**
Throttle	**1,800 RPM**
Magnetos	**CHECK (max drop 150 RPM, max 50 RPM differential)**
Engine instruments / ammeter	**CHECK (oil press 50–90 psi green)**
Vacuum gauge (analogue)	**CHECK (4.5–5.5 in Hg)**
Annunciators	**CHECK (none lit)**
Throttle	**1,000 RPM**
Throttle friction lock	**ADJUST**
Autopilot	**OFF**
Radios / transponder	**SET / ALT**
Strobe lights	**ON**
Wing flaps	**0°–10°**
Takeoff briefing	**COMPLETE**

After Takeoff / Climb

Airspeed	**74 KIAS (Vy) / 62 KIAS (Vx, obstacles)**
Wing flaps	**UP (above 60 KIAS)**
Throttle	**FULL OPEN**
Mixture	**RICH (lean above 3,000 ft for smooth operation)**
Engine instruments	**CHECK**
Fuel pump	**OFF**
Landing / taxi lights	**OFF**

Cruise

Power	**2,100–2,700 RPM (80% max continuous)**
Elevator trim	**ADJUST**
Mixture	**LEAN as required (per fuel flow / EGT)**
Engine instruments	**CHECK**
Fuel quantity / endurance	**CHECK**
Heading indicator (analogue)	**ALIGN with compass**

Descent

Altimeter / barometric setting	**SET**
Power	**AS DESIRED**
Mixture	**ADJUST for smooth operation**
Fuel selector	**BOTH**
Approach briefing	**COMPLETE**

Before Landing

Seats, belts, harnesses	**SECURE**
Fuel selector	**BOTH**
Mixture	**RICH**
Landing / taxi lights	**ON**
Autopilot	**OFF**
Airspeed	**65–75 KIAS (flaps up)**
Wing flaps	**AS DESIRED (10° below 110 KIAS, 10°–30° below 85 KIAS)**
Airspeed	**60–70 KIAS (flaps 30°)**

After Landing

Wing flaps	**UP**
Strobe / landing lights	**OFF / AS REQUIRED**
Transponder	**STBY / as required**

Shutdown

Parking brake	**SET**
Throttle	**IDLE (1,000 RPM then idle)**
Electrical equipment / autopilot / pitot heat	**OFF**
Avionics switch (BUS 1 & BUS 2 on G1000)	**OFF**
Mixture	**IDLE CUTOFF (pull full out)**
Ignition switch	**OFF (key removed)**
Master switch (ALT & BAT)	**OFF**
Standby battery	**OFF (G1000)**
Control lock	**INSTALL**
Fuel selector	**LEFT or RIGHT (prevents crossfeed on sloped ramp)**

Abnormal procedures

LOW VOLTS / Alternator Failure

LOW VOLTS annunciator or ammeter discharge in flight

Avionics switch (BUS 1 & BUS 2)	**OFF**
Master switch (ALT only)	**OFF, then ON**
LOW VOLTS annunciator / ammeter	**CHECK**
Avionics switch (BUS 1 & BUS 2)	**ON**

▶ Charging restored?

Yes:

Ammeter / volts	**MONITOR**

No:

Master switch (ALT only)	**OFF**
Non-essential electrical equipment	**OFF (lights, autopilot, BUS 2 avionics)**
Standby battery	**CHECK ARMED – approx 30 min essential bus (G1000)**
Deferred: Alternator failed – battery / standby battery only, limited endurance. Land as soon as practical, lower flaps early while power remains.	**NOTED**

Low Oil Pressure

OIL PRESSURE annunciator or pressure below 50 psi green arc

Oil pressure / oil temperature	**CHECK**

▶ Oil temperature also rising?

Yes:

Power	**REDUCE to minimum required**
Engine failure	**EXPECT – plan a precautionary landing**
Deferred: Low oil pressure with high temperature – land as soon as possible, expect engine failure.	**NOTED**

No – temp normal:

Flight	**CONTINUE to nearest suitable airport, monitor**
Deferred: Low oil pressure indication – land as soon as practical, maintenance check.	**NOTED**

High Oil Temperature

Oil temperature above 245°F / in red arc

Mixture	**RICH**
Power	**REDUCE**
Airspeed	**INCREASE (lower nose for cooling)**
Oil pressure	**MONITOR**

▶ Temperature returning to green arc?

Yes:

Flight	**CONTINUE, monitor**

No:

Deferred: High oil temperature – land as soon as practical, minimum power.	**NOTED**

Rough Running / Fuel Flow Fluctuation

Rough engine, fuel flow fluctuating (vapour) or power loss

Fuel pump	**ON**
Mixture	**ADJUST for smooth operation**
Fuel selector	**BOTH**

▶ Running smooth?

Yes:

Fuel pump	**OFF when fuel flow stable (ON again if fluctuation returns)**

No:

Magnetos	**CHECK L / R / BOTH, select smoothest**
Alternate induction air	**OPENS AUTOMATICALLY (no action)**
Deferred: Rough running engine – land as soon as practical, be ready for engine failure.	**NOTED**

Flap Failure

Flap circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Approach ..... **FLAPS UP – 65–75 KIAS, flatter approach, longer runway**

Deferred: Flaps inoperative – flaps-up landing at 65–75 KIAS, increased landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **REDUCE (approx 75 KIAS)**

Aircraft ..... **CONTROL – door has little effect on handling**

Window on open-door side ..... **OPEN (equalise pressure), then door CLOSE & LATCH**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Deferred: Door unlatched – land as soon as practical, normal landing, keep speed low. ..... **NOTED**

Vacuum / AHRS / PFD Failure

Vacuum gauge low or gyro flags (analogue); red X on PFD or display blank (G1000)

▶ Panel type?

G1000:

Display backup (red button) ..... **PRESS – reversionary mode on remaining display**

Standby airspeed / attitude / altimeter ..... **USE and cross-check**

Standby battery ..... **CHECK ARMED**

Deferred: G1000 display / AHRS failed – reversionary mode or standby instruments; avoid IMC. ..... **NOTED**

Analogue:

Vacuum gauge ..... **CHECK (4.5–5.5 in Hg)**

Attitude / heading indicators ..... **DISREGARD – cross-check**

Turn coordinator, compass, altimeter, airspeed ..... **USE for partial-panel flight**

Deferred: Vacuum failed – attitude and heading indicators unreliable; VMC only. ..... **NOTED**

IMC ..... **AVOID / EXIT**

Inadvertent Icing Encounter

Pitot heat ..... **ON**

Exit icing conditions ..... **TURN BACK / CHANGE ALTITUDE**

Cabin heat / windshield defrost ..... **FULL ON**

Power ..... **INCREASE to minimise ice on propeller**

Mixture ..... **ADJUST (alternate induction air opens automatically)**

Airspeed on approach ..... **INCREASE (flaps up or 10° max with residual ice)**

Deferred: Residual airframe ice – flaps-up / 10° landing, approach 10–15 KIAS faster, expect higher stall speed. ..... **NOTED**

Static Source Blocked

Erroneous airspeed / altimeter / VSI

Alternate static valve ..... **PULL ON**

Cabin vents / windows ..... **CLOSED when alternate static in use**

Airspeed / altimeter ..... **APPLY POH alternate static corrections**

Deferred: Alternate static in use – airspeed and altimeter read slightly high; add margin on approach. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

On the runway

Throttle (memory) ..... **IDLE**

Brakes (memory) ..... **APPLY**

Wing flaps (memory) ..... **RETRACT**

Mixture ..... **IDLE CUTOFF**

Ignition switch ..... **OFF**

Master switch (ALT & BAT) ..... **OFF**

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	70 KIAS flaps up / 65 KIAS flaps 10°–30°
Mixture (memory)	IDLE CUTOFF
Fuel shutoff valve (memory)	OFF (pull full out)
Ignition switch (memory)	OFF
Wing flaps (memory)	AS REQUIRED (30° recommended)
Master switch (ALT & BAT) (memory)	OFF
Cabin door	UNLATCH

Land straight ahead with only small changes of direction. Do not attempt to turn back to the runway.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	68 KIAS (best glide)
Fuel shutoff valve (memory)	ON (push full in)
Fuel selector (memory)	BOTH
Fuel pump (memory)	ON
Mixture (memory)	RICH (if restart not immediate: IDLE CUTOFF, then advance slowly)
Ignition switch (memory)	BOTH (START if propeller stopped)

▶ Engine restarted?

Yes:

Power	AS REQUIRED
Fuel pump	OFF (ON if power fluctuates)
Engine instruments	MONITOR
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system or ignition.	NOTED

No – forced landing:

Forced landing without engine power:

Field / wind	SELECT & PLAN (into wind if possible)
Mayday / 7700	TRANSMIT (time permitting)
Seats, belts, harnesses	SECURE
Airspeed	70 KIAS flaps up / 65 KIAS flaps down
Mixture	IDLE CUTOFF
Fuel shutoff valve	OFF (pull full out)
Ignition switch	OFF
Wing flaps	AS REQUIRED (30° recommended)
Master switch (ALT & BAT)	OFF (when landing assured)
Doors	UNLATCH prior to touchdown
Touchdown	SLIGHTLY TAIL LOW
Brakes	APPLY HEAVILY

Engine Fire During Start

Cranking (memory)	CONTINUE (draw flames into engine)

▶ Engine started?

Yes:

Power	1,800 RPM for a few minutes
Engine	SHUT DOWN & INSPECT for damage

No:

Throttle	FULL OPEN
Mixture	IDLE CUTOFF
Cranking	CONTINUE
Fuel pump	OFF
Fuel shutoff valve	OFF
Ignition switch	OFF
Master switch (ALT & BAT)	OFF
Fire extinguisher	OBTAIN & USE
Aircraft	EVACUATE, inspect for damage

Engine Fire in Flight

LAND IMMEDIATELY

- Mixture (memory) ..... **IDLE CUTOFF**
- Fuel shutoff valve (memory) ..... **OFF (pull full out)**
- Fuel pump (memory) ..... **OFF**
- Master switch (ALT & BAT) (memory) ..... **OFF**
- Cabin heat / air (memory) ..... **OFF (except overhead vents)**
- Airspeed (memory) ..... **100 KIAS (increase if fire persists until incombustible)**

Do not attempt an engine restart.

- Forced landing ..... **EXECUTE (see Engine Failure in Flight)**

Electrical Fire in Flight

LAND AS SOON AS POSSIBLE

- Standby battery (memory) ..... **OFF (G1000)**
- Master switch (ALT & BAT) (memory) ..... **OFF**
- Vents / cabin air / heat (memory) ..... **CLOSED**
- Fire extinguisher ..... **ACTIVATE (if available)**
- Avionics switch (BUS 1 & BUS 2) ..... **OFF**
- All other switches (except ignition) ..... **OFF**

After discharging the extinguisher, ventilate the cabin.

▶ Fire out and electrical power necessary?

Yes:

- Master switch (ALT & BAT) ..... **ON**
- Standby battery ..... **ARM (G1000)**
- Circuit breakers ..... **CHECK – do not reset a faulted breaker**
- Avionics switch (BUS 1 then BUS 2) ..... **ON one at a time, with delay, check for smoke**
- Vents / cabin air / heat ..... **OPEN when fire is completely out**
- Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. ..... **NOTED**

No:

Continue with master off. No flaps, radios or fuel gauges; standby instruments only (G1000).

- Deferred: Electrical power off – flaps-up landing, no radio; standby instruments only. ..... **NOTED**

Cabin Fire

LAND AS SOON AS POSSIBLE

- Master switch (ALT & BAT) (memory) ..... **OFF**
- Vents / cabin air / heat (memory) ..... **CLOSED**
- Fire extinguisher ..... **ACTIVATE**
- Cabin ..... **VENTILATE after fire is out**
- Land ..... **AS SOON AS POSSIBLE, inspect for damage**

Emergency Descent

- Throttle (memory) ..... **IDLE**
- Mixture (memory) ..... **RICH**
- Wing flaps (memory) ..... **UP**
- Airspeed (memory) ..... **129 KIAS smooth air / 105 KIAS turbulence (Va)**
- Power ..... **CLEAR ENGINE periodically**

Spin Recovery

Inadvertent spin

- Ailerons (memory) ..... **NEUTRAL**
- Throttle (memory) ..... **IDLE**
- Rudder (memory) ..... **FULL OPPOSITE to direction of rotation**
- Control wheel (memory) ..... **BRISKLY FORWARD (just after rudder reaches stop)**
- Controls (memory) ..... **HOLD until rotation stops**
- Rudder (memory) ..... **NEUTRALISE as rotation stops**
- Recovery ..... **SMOOTH pull-out from the dive**

Flaps must be retracted; if extended, retract immediately after recovery.